

# **Application of Information & Communication Technologies**

## **Lecture-10**

# Recap of Lecture 9

- ◆ Input Devices
- ◆ How humans give data and instructions to the computer

# Overview of Lecture 10

- ◆ Output Devices
- ◆ How computers give results back to us

# How computers give results back to us



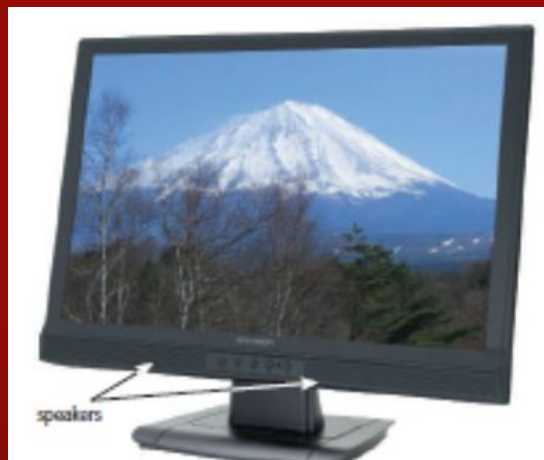
# What are Output Devices?

- ◆ Devices that show results of computer processing.
- ◆ Convert binary data → human-readable form.
- ◆ Examples: Monitor, Printer, Speakers.

# Monitors

- ◆ Main output device for visuals.
- ◆ Show text, images, videos.
- ◆ Measured in inches & resolution (HD, 4K).
- ◆ Types of Monitors
  - CRT (old) – heavy, box-shaped.
  - LCD/LED – thin, flat, energy efficient.
  - OLED – very sharp colors, flexible screens.

# Monitors



# Data and Multimedia Projectors

- ◆ **Data projector:** Display device that projects all computer output to a wall or projection screen
  - Found in classrooms, conference rooms
  - Can be wireless or integrated into devices
  - Integrated – tiny projectors built into mobile phones, portable devices
  - Hologram projectors used to display 3D images



**FIGURE 4-34**

Data projectors.



**CONVENTIONAL DATA PROJECTORS**  
Frequently used for both business and classroom presentations.



**INTEGRATED AND PORTABLE PROJECTORS**  
Images displayed on the device (such as the mobile phone shown here) are projected onto any surface.



**HOLOGRAPHIC PROJECTORS**  
Project 3D images of objects or individuals.



# Audio Output

- ◆ **Audio output:** Output in the form of voice, music, and other audible sounds
  - **Speakers :** Convert digital audio → sound.
  - **Headphones and headsets:** Personal listening. Wired / Wireless (Bluetooth).
  - **Earphones and earbuds**
- ◆ **Special Audio Systems: Voice output systems (for blind).**
  - **Smart assistants (Alexa, Google Home).**



**COMPUTER SPEAKERS**

Used to output sound from a computer.



**IPOD/MP3 DOCK**

Used to output sound from a portable digital media player.



**HEADSETS**

Used when both voice input and audio output are required.

**FIGURE 4-41**  
Audio output devices.

# Printers Basics

- ◆ Produce hard copy (paper output).
- ◆ Speed = pages per minute (ppm).
- ◆ Quality = dots per inch (dpi).



**FIGURE 4-35**  
**Dot-matrix printers.**  
Dot-matrix printers are impact printers; today they are typically high-speed printers used in manufacturing, shipping, or similar applications.

# Impact & non-Impact Printers

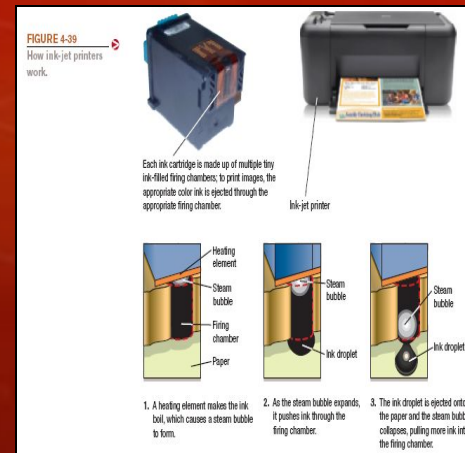
## Impact Printers

- ◆ Dot Matrix → uses pins + ink ribbon.
- ◆ Noisy, low quality.
- ◆ Still used for bills, receipts.



## Non-Impact Printers

- ◆ Inkjet → sprays ink, good quality, cheap.
- ◆ Laser → toner + drum, fast & sharp.



# Special-Purpose Printers

- ◆ Photo printers
- ◆ Barcode, label, and postage printers
- ◆ Portable printers
- ◆ Plotters and wide-format ink-jet printers
- ◆ 3-D printers



# Plotters

- ◆ Print large drawings (maps, blueprints).
- ◆ Used by engineers & architects.



# VR Headsets

- ◆ Show 3D virtual worlds.
- ◆ Used in gaming, training, simulations.





# Braille Displays

- ◆ For blind users.
- ◆ Convert text → raised dots (touch reading).



# Summary Table

| Device     | Output Type | Example Use  |
|------------|-------------|--------------|
| Monitor    | Visual      | Work, movies |
| Speakers   | Audio       | Music        |
| Printer    | Hard copy   | Reports      |
| VR Headset | Immersive   | Gaming       |



# Activity (Class)

- ◆  Discuss:
  - Which output device do you use most daily?
  - Would you survive with only sound output? Why?

# Quick Quiz

- ◆ Which printer is best for high-quality photos?
- ◆ What is the main difference between Inkjet & Laser printer?
- ◆ Name one output device for blind people.

# Future of Output Devices

- ◆ 8K displays & holographic screens.
- ◆ 3D & 4D printing.
- ◆ Brain-computer interface outputs.

# Summary

- ◆ Output devices = How computers “talk back” to us.

# Suggested Reading

- ◆ Ch-04, The System Unit: Processing and Memory ,  
“Understanding Computers: Today and Tomorrow,  
Comprehensive”, 15th Edition by Deborah Morley  
& Charles S. Parker
- ◆ Ch-05, Discovering Computers Fundamentals-  
Your Interactive Guide -- Gary B Shelly; Misty E  
Vermaat; Jeffrey J Q